

BRAD CAO

(301) 356-1069 | bcao4049@gmail.com | [LinkedIn](#) | github.com/bradca0

EDUCATION

Columbia University

B.A. in Mathematics & Computer Science

New York, NY

Expected May 2028

Relevant Coursework: Data Structures in Java, Probability Theory, Natural Language Processing, Honors Mathematics A & B

EXPERIENCE

Data Science Intern

Mercor

Apr. 2026 – Present

San Francisco, CA

- Engineered evaluation frameworks that formalize subject-matter expertise into RLHF-compatible annotation schemas for LLM fine-tuning pipelines
- Designed rubric-based scoring taxonomies and QA workflows to raise inter-annotator agreement; audited quality and flagged edge cases to refine labeling guidelines

Research Assistant

Duke University

Apr. 2026 – Present

Durham, NC

- Trained and developed LLMs to identify and quantify inter-network disparagement in televised news media, building NLP pipelines to ingest, segment, and label large transcript volumes
- Benchmarked prompt-engineered and fine-tuned classifiers against human-coded samples with Dr. Philip Napoli, translating media-studies frameworks into quantifiable schemas

Research Intern

University of Notre Dame

Jun. 2025 – Aug. 2025

South Bend, IN

- Awarded a full-ride scholarship for original quantitative research on the pharmaceutical industry's impact on public health policy
- Applied R-based regression and visualization to link industry practices to poverty outcomes; presented at a symposium

Research Intern

University of Maryland, Baltimore

Jun. 2023 – Sep. 2025

Baltimore, MD

- First-authored a supervised ML model with Prof. Allison Dunn predicting ADHD-related patient safety risks from real-world data
- First-authored a study modeling tacrolimus dose requirements and transplant outcomes in lung transplant recipients on systemic azole antifungals
- Presented at the ASCPT Annual Meetings (2024, 2025); earned the Top Poster Award (May 2025)

PROJECTS

TrueFill | Python, NumPy, scikit-learn | github.com/bradca0/lobsim

- Built an event-driven limit-order-book simulator with exact queue-position tracking; a market-making strategy worth +214 ticks/episode under standard fill assumptions falls to +2.8 once queue position is modeled (−98.7%, 200 held-out episodes)
- Calibrated a synthetic market (Hawkes + latent fundamental) to 9 of 11 published stylized facts; trained a Fitted Q-Iteration market maker over GBDTs with an 890× inference speedup
- Validated with paired bootstrap CIs, Holm–Bonferroni, and deflated Sharpe; 263 tests, 95.7% coverage, `mypy --strict`, one-command reproducible

PolyEdge | Python | github.com/bradca0/PolyEdge

- Measured skill vs. variance among Polymarket traders by reconstructing every trader's realized P&L from raw on-chain data (six event types absent from the trade feed), validated against the exchange's position data: 92% of positions match exactly (829/901, 51 wallets); surfaced four silent API bugs
- Modeled trader skill with James–Stein shrinkage — a raw population edge of +0.126 collapses to −0.095, showing 62% of the cross-wallet spread is sampling noise
- Point-in-time backtest with costs shows skill-weighted forecasts do not beat market price (Brier 0.209 vs. 0.205) — a credible negative result; 86 tests, `ruff + mypy --strict`

MyCureHope | Django, PostgreSQL, JavaScript, TailwindCSS | mycurehope.org

- Co-developed an open-source patient database of treatments, doctors, and hospitals serving 33,000+ users across 78 countries; own search build and technical development
- Built and deployed the full-stack platform (Django, PostgreSQL, JavaScript, TailwindCSS) on Nginx/Gunicorn/Ubuntu

PUBLICATIONS

Cao, B., Lee, S., Naga, A., Gobburu, J., Dunn, A. “Leveraging Supervised Machine Learning to Identify Predictors of Serious Clinical Outcome with Stimulant Treatment from Real-World Data.” *ASCPT Annual Meeting*, 2025.

Cao, B., Gupta, P., Dunn, A. “Predictive Modeling of Tacrolimus Dose Requirements and Transplantation Outcomes in Lung Transplant Recipients on Systemic Azole Antifungals.” *ASCPT Annual Meeting*, 2024.

TECHNICAL SKILLS & INTERESTS

Languages: Python, JavaScript, Java, SQL (PostgreSQL), R, HTML/CSS

Frameworks & Tools: Django, TailwindCSS, NumPy, pandas, scikit-learn, Nginx, Gunicorn, Ubuntu/Linux

ML & Data: LLM fine-tuning, RLHF annotation, supervised ML, data analysis

Interests: Fencing (USA Fencing First Team All-American), Competitive Programming (USACO), History Olympiads, Open-Source